

Data Flow Diagram

Date	07-07-2026
Team ID	
Project Name	Automated Credit Card Approval Prediction using Machine Learning
Maximum Marks	2 Marks

Data Flow Diagram

A **Data Flow Diagram (DFD)** illustrates how applicant information flows through the **Credit Card Approval Prediction System**, from data submission to prediction and result display.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person or system outside the application that sends or receives data (e.g., Applicant).
Rectangle with numbered header	Process	A process that transforms input into output (e.g., Validate Input, Predict Approval).
Rectangle (open, no number)	Data Store	Stores trained machine learning model and preprocessing files.
Labeled arrow	Data Flow	Represents the movement of data between entities, processes, and data stores.

Level 0 Data Flow Diagram

Project: Automated Credit Card Approval Prediction using Machine Learning

Main Process:

- 1.0 Input Collection — Inputs include Age, Income, Employment Status, Credit History, Loan Amount, Existing Debt.
- 2.0 Validate & Scale Input — Handle missing values, encode categorical data, and apply StandardScaler.
- 3.0 Predict Crop — Use the trained Machine Learning model (Logistic Regression/Random Forest/XGBoost) to predict whether the application is **Approved** or **Rejected**.
- 4.0 Display Result — Display the approval prediction on the result page.

External Entity:

Applicant: Provides personal and financial information.

Receives the approval or rejection result.

Data Store

D1: Trained ML Model

Stores the trained Machine Learning model (credit_card_model.pkl)

Stores preprocessing files (scaler.pkl, encoder.pkl)

Data Flows

Applicant → 1.0 Input Collection : Personal & Financial Details

1.0 Input Collection → 2.0 Validate & Preprocess Input : Raw Applicant Data

2.0 Validate & Preprocess Input → 3.0 Predict Credit Card Approval : Processed Feature Vector

D1 → 3.0 Predict Credit Card Approval : Load Trained Model & Scaler

3.0 Predict Credit Card Approval → 4.0 Display Result : Approval / Rejection Prediction

4.0 Display Result → Applicant : Credit Card Approval Status

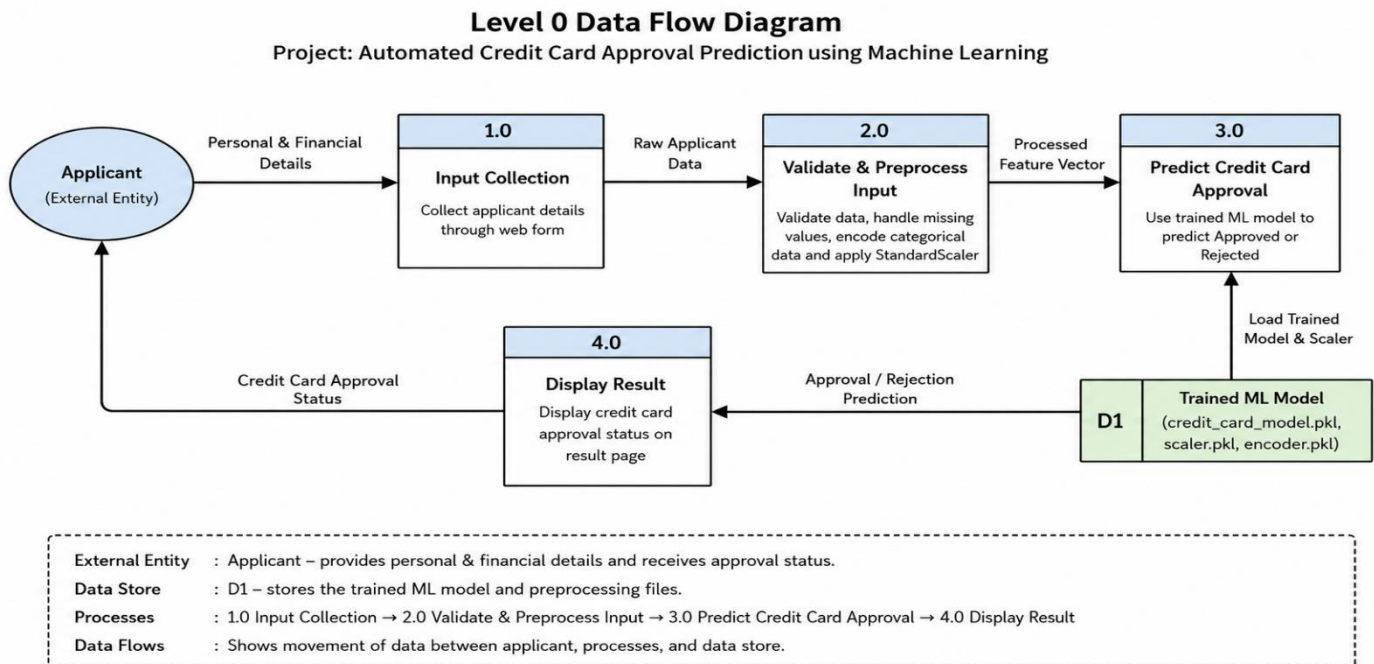


Figure 1: Level 0 DFD for *Automated Credit Card Approval Prediction using Machine Learning*